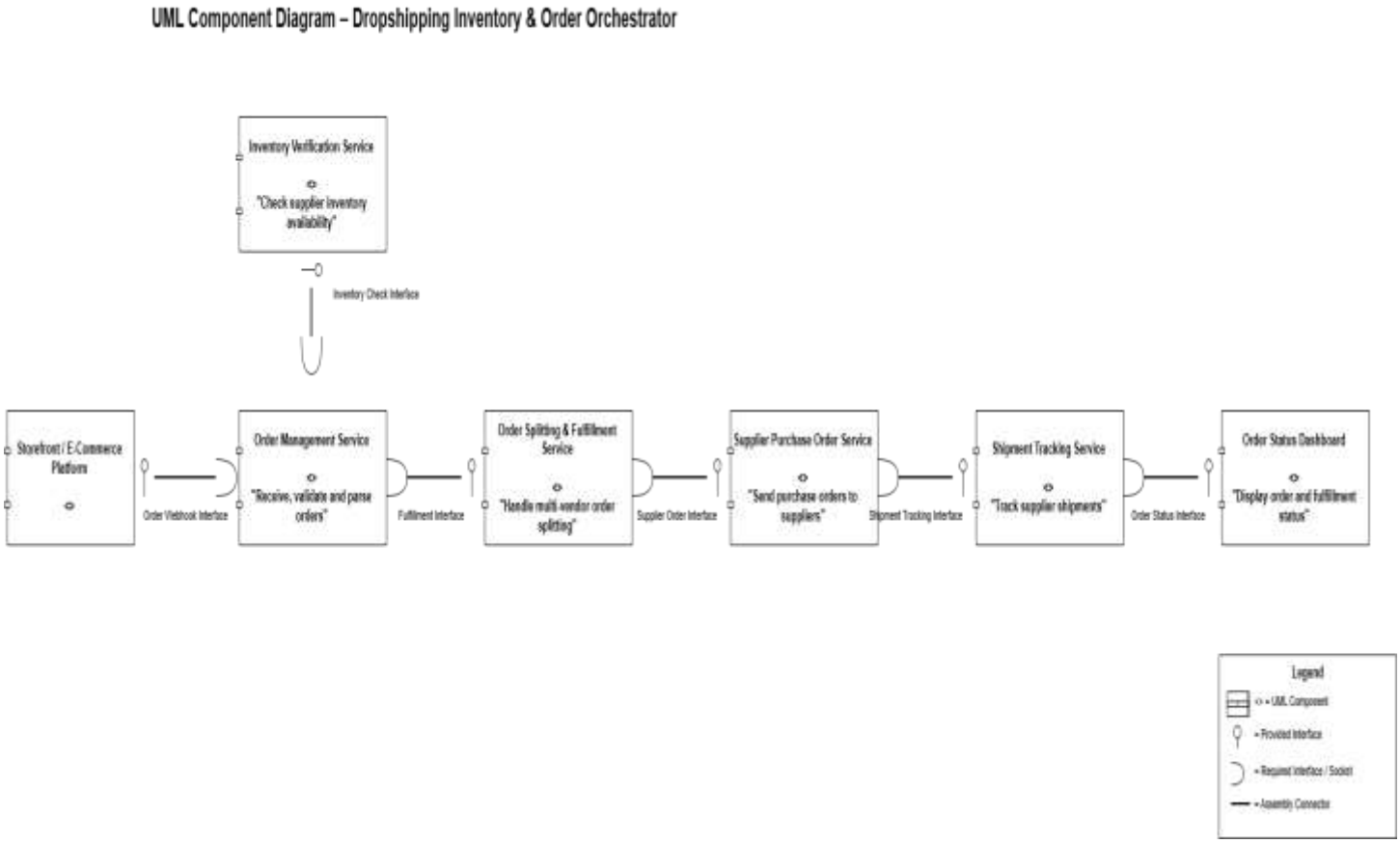
SOFTWARE ENGINEERING – LAB 3

Component Modelling & Architectural Pattern Selection

Dropshipping Inventory & Order Orchestrator

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Section	B	**Date**	9-9-2026

1. UML Component Diagram



2. Written Justification

Architectural Choice

We chose Microservices Architecture for the Dropshipping Inventory & Order Orchestrator System. The system is divided into independent components such as Order Management, Inventory Verification, Order Splitting & Fulfillment, Supplier Purchase Order, Shipment Tracking, and Order Status Dashboard.

Reason 1 – Independent Service Scaling

Different parts of the dropshipping workflow can experience different workloads. For example, inventory verification and order processing may receive many requests at the same time. With microservices, individual services can be scaled independently without scaling the entire application.

Reason 2 – Fault Isolation and Independent Deployment

The system integrates with storefronts and multiple suppliers, so failures or changes in one area should not require

the whole system to be changed. Separating order processing, supplier purchase orders, and shipment tracking into services provides better fault isolation and allows individual services to be updated independently.

Security Advantage
Microservices allow security responsibilities to be separated between services. For example, webhook validation can be handled by the Order Management Service, while supplier integrations can be isolated within the Supplier Purchase Order Service. This reduces unnecessary exposure between different parts of the system and supports service-level access control.

Performance Benefit
The architecture can improve performance by allowing frequently used services to scale independently and by keeping each service focused on a specific responsibility. Inventory checks, order splitting, supplier communication, and shipment tracking can therefore operate without requiring a single large application to handle every operation.

Conclusion
Microservices is appropriate for the Dropshipping Inventory & Order Orchestrator because the workflow contains several distinct functions and external supplier/storefront integrations. The selected component structure supports separation of responsibilities, independent scaling, fault isolation, security, and maintainability.